endee-io / **endee**

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Watch 4

Endee.io – A high-performance vector database, designed to handle up to 1B vectors on a single node, delivering significant performance gains through optimized indexing and execution. Also available in cloud https://endee.io/

endee.io

Apache-2.0 license

Contributing

683 stars 919 forks 4 watching Branches Activity Custom properties Tags

Public repository

32 Branches 3 Tags Go to file t Go to file Add file Code

shaleenji feature: Weighted rank fusion start (#88) 267a750 · 7 hours ago

docs	restructuring getting_started docs (...	last week
infra	start	2 months ago
src	feature: Weighted rank fusion start (...	7 hours ago
tests	Sparse algo bmw (#49)	3 weeks ago
third_party	Sparse performance enhancement (...	last week
.clang-format	start	2 months ago
.gitignore	Filter design (#32)	last month
CMakeLists.txt	Sparse performance enhancement (...	last week
CONTRIBUTING.md	License (#4)	2 months ago
LICENSE	License (#4)	2 months ago
README.md	readme update	last week
docker-compose.yml	start	2 months ago
install.sh	refactor: bump web ui version to 1.2...	last week
run.sh	ndd binary soft link instead of differ...	last month

README | Contributing | Apache-2.0 license

High-performance open-source vector database for AI search, RAG, semantic search, and hybrid retrieval.

Quick Start Local Setup | Docs Quick Start | license Apache-2.0 | Discord Join Chat | Website Endee

Quick Start • Why Endee • Use Cases • Features • API and Clients • Docs • Contact

Endee: Open-Source Vector Database for AI Search

Endee is a high-performance open-source vector database built for AI search and retrieval workloads. It is designed for teams building **RAG pipelines**, **semantic search**, **hybrid search**, recommendation systems, and filtered vector retrieval APIs that need production-oriented performance and control.

Endee combines vector search with filtering, sparse retrieval support, backup workflows, and deployment flexibility across local builds and Docker-based environments. The project is implemented in C++ and optimized for modern CPU targets, including AVX2, AVX512, NEON, and SVE2.

If you want the fastest path to evaluate Endee locally, start with the Getting Started guide or the hosted docs at docs.endee.io.

Why Endee

- Built as a dedicated vector database for AI applications, search systems, and retrieval-heavy workloads.
- Supports dense vector retrieval plus sparse search capabilities for hybrid search use cases.
- Includes payload filtering for metadata-aware retrieval and application-specific query logic.
- Ships with operational features already documented in this repo, including backup flows and runtime observability.
- Offers flexible deployment paths: local scripts, manual builds, Docker images, and prebuilt registry images.

Getting Started

The full installation, build, Docker, runtime, and authentication instructions are in docs/getting-started.md.

Fastest local path:

```
chmod +x ./install.sh ./run.sh
./install.sh --release --avx2
./run.sh
```

The server listens on port `8080`. For detailed setup paths, supported operating systems, CPU optimization flags, Docker usage, and authentication examples, use:

- Getting Started
- Hosted Quick Start Docs

Use Cases

RAG and AI Retrieval

Use Endee as the retrieval layer for question answering, chat assistants, copilots, and other RAG applications that need fast vector search with metadata-aware filtering.

Agentic AI and AI Agent Memory

Use Endee as the long-term memory and context retrieval layer for AI agents built with frameworks like LangChain, CrewAI, AutoGen, and LlamaIndex. Store and retrieve past observations, tool outputs, conversation history, and domain knowledge mid-execution with low-latency filtered vector search, so your autonomous agents get the right context without stalling their reasoning loop.

Semantic Search

Build semantic search experiences for documents, products, support content, and knowledge bases using vector similarity search instead of exact keyword-only matching.

Hybrid Search

Combine dense retrieval, sparse vectors, and filtering to improve relevance for search workflows where both semantic understanding and term-level precision matter.

Recommendations and Matching

Support recommendation, similarity matching, and nearest-neighbor retrieval workflows across text, embeddings, and other high-dimensional representations.

Features

- **Vector search** for AI retrieval and semantic similarity workloads.
- **Hybrid retrieval support** with sparse vector capabilities documented in docs/sparse.md.
- **Payload filtering** for structured retrieval logic documented in docs/filter.md.
- **Backup APIs and flows** documented in docs/backup-system.md.
- **Operational logging and instrumentation** documented in docs/logs.md and docs/mdbx-instrumentation.md.
- **CPU-targeted builds** for AVX2, AVX512, NEON, and SVE2 deployments.
- **Docker deployment options** for local and server environments.

API and Clients

Endee exposes an HTTP API for managing indexes and serving retrieval workloads. The current repo documentation and examples focus on running the server directly and calling its API endpoints.

Current developer entry points:

- Getting Started for local build and run flows
- Hosted Docs for product documentation
- Release Notes 1.0.0 for recent platform changes

Docs and Links

- Getting Started
- Hosted Documentation
- Release Notes
- Sparse Search
- Filtering
- Backups

Community and Contact

- Join the community on Discord
- Visit the website at endee.io
- For trademark or branding permissions, contact enterprise@endee.io

Contributing

We welcome contributions from the community to help make vector search faster and more accessible for everyone.

- Submit pull requests for fixes, features, and improvements
- Report bugs or performance issues through GitHub issues
- Propose enhancements for search quality, performance, and deployment workflows

License

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Releases 3

v1.1.0 Latest
7 hours ago

+ 2 releases

Packages

No packages published

Contributors 13

Languages

- C++ 96.5%
- CMake 1.7%
- Shell 1.5%
- Dockerfile 0.3%